A LOGARITHMIC IMPROVEMENT OF VAUGHAN'S BOUND FOR THE ERDŐS–STRAUS EXCEPTIONAL SET

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Abstract. Let $E(N)$ be the number of $n \le N$ for which $4/n$ is not a sum of three unit fractions. Vaughan (1970) proved $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$. We show

$$E(N) \ll N \exp\{-c(\log N)^{2/3}(\log\log N)^{1/3}\}.$$

The proof is Vaughan's large-sieve argument applied to the residue classes modulo $k\ell$ (rather than modulo ℓ alone) that force solvability, where ℓ is a prime and $k \le \delta \log N$ a "multiplier", after splitting the sieved set into progressions modulo $\operatorname{lcm}\{k\}$. The new ingredient is a count of the forced classes, based on an elementary lattice estimate, which is uniform in the progression. Only Bombieri–Vinogradov, Brun–Titchmarsh and Montgomery's large sieve are used.

1. Introduction

The Erdős–Straus conjecture asserts that for every integer $n \ge 2$ there are positive integers x, y, z with

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

It has been verified for $n \le 10^{18}$ (see [5], §6, and [6] for 10^{17}), and Mordell's identities show that it holds unless $n \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$ [4]. Let

$$E(N) = \#\{n \le N : \ (1) \text{ has no solution}\},$$
$$E_{\mathrm{pr}}(N) = \#\{p \le N \text{ prime} : \ (1) \text{ has no solution}\}.$$

Vaughan [7] proved that $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$ for an absolute $c > 0$; a self-contained account of his argument, uniform in the numerator, is given by Pomerance and Weingartner [5], whose Theorem 1.3 retains this exponent for fixed numerators. We prove:

Theorem 1.1. *There is an absolute constant $c > 0$ such that for all $N \ge 3$*

$$E_{\mathrm{pr}}(N) \ll N \exp\{-c(\log N)^{2/3}(\log\log N)^{1/3}\}.$$

Theorem 1.2. *The same bound holds for $E(N)$.*

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The exponent $2/3$ is unchanged; the gain is the factor $(\log\log N)^{1/3}$ inside the exponential. We briefly recall where Vaughan's $2/3$ comes from, because the improvement is located precisely there. For a prime $\ell \equiv 3 \pmod 4$, every n in one of roughly $\tau((\ell+1)^2/16)$ explicit residue classes modulo ℓ satisfies (1); on average over ℓ this is $\asymp (\log \ell)^2$ classes, so $\sum_{\ell \le X} \omega(\ell)/\ell \asymp (\log X)^2$. Feeding these classes into Montgomery's large sieve saves a factor $\exp\{c(\log X)^2\}$, and the sieve level constraint forces $\log X \asymp (\log N)^{1/3}$.

The observation here is that for each "multiplier" $k \equiv 1 \pmod 4$ the number $(k\ell+1)/4$ also produces forced classes, now modulo $k\ell$ (Lemma 2.1). Restrict n to a fixed reduced progression modulo $L = \operatorname{lcm}\{k \le K,\ k \equiv 1\ (4)\}$. The compatible classes become classes modulo ℓ; the small coprime factorizations selected in Lemma 4.1 give distinct classes, even for different k. Their averaged contribution multiplies the class mass by

$$h(K) = \sum_{\substack{k \le K \\ k \equiv 1\ (4)}} \frac{\varphi(k)}{k^2} \asymp \log K.$$

The price is that the large sieve is applied to an interval of length N/L; we choose $L \le N^{1/2}$, ensured by taking K a small constant times $\log N$. Hence the gain $\log K \asymp \log\log N$ and, after re-optimising, the exponent $(\log N)^{2/3}(\log\log N)^{1/3}$. Since

$$\sum_{k \mid L} \frac{\varphi(k)}{k^2} \le \prod_{p \mid L} (1 - 1/p)^{-1} \ll \log\log(3L),$$

no multiplier family with $L \le N^{1/2}$ improves the order of this particular weighted multiplier sum beyond $\log\log N$.

The argument is short. Sections 2–4 construct and count the forced classes; Section 5 runs the large sieve; Section 6 deduces Theorem 1.2 from Theorem 1.1 by a Rankin argument on the semigroup generated by exceptional primes. Throughout, p and ℓ denote primes, τ is the divisor function, φ Euler's function, and all constants are absolute unless indicated.

2. Forced residue classes

Lemma 2.1 (multiplier identity)**.** *Let k, ℓ be positive integers with $k\ell \equiv 3 \pmod 4$, and let $A = (k\ell+1)/4 = uvw$ be any factorization into positive integers. If $n \ge 1$ satisfies $nv \equiv -u \pmod{k\ell}$, then $s = (nv+u)/(k\ell)$ is a positive integer and*

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw}.$$

In particular (1) is solvable for every n in the residue class $-uv^{-1} \pmod{k\ell}$ (note $(v, k\ell) = 1$ since $(A, k\ell) = 1$).

Proof. On the common denominator $nsuvw$ the three numerators sum to $nv + u + s = sk\ell + s = s(k\ell+1) = 4suvw$. □

For $k = 1$ these are the classes used by Vaughan. We shall use primes $\ell \equiv 3 \pmod 4$ and multipliers $k \equiv 1 \pmod 4$. The condition $nv \equiv -u \pmod{k\ell}$ splits, when $(k,\ell) = 1$, into

$$n \equiv -uv^{-1} \pmod \ell \qquad \text{and} \qquad u + nv \equiv 0 \pmod k. \tag{2}$$

The second condition depends on n only modulo k. Fix $K \ge 1$ and put

$$\mathcal{K}(K) = \{k \le K :\ k \equiv 1\ (4)\}, \qquad L_K = \operatorname{lcm} \mathcal{K}(K),$$

$$h(K) = \sum_{k \in \mathcal{K}(K)} \frac{\varphi(k)}{k^2}.$$

If $n \equiv c \pmod{L_K}$, the second condition in (2) reads $u + cv \equiv 0 \pmod k$ and is a condition on (u, v, k) alone; then every n in the progression $c \pmod{L_K}$ lying in the class $-uv^{-1} \pmod \ell$ is solvable.

3. Lattice sums

For $k \ge 1$, $(c,k) = 1$ and $1 \le H < z$ let

$$\mathcal{S}(k, c; H, z) = \{(u,v) :\ H < u, v \le z,\ (u,v) = 1,\ (v,k) = 1,\ u \equiv -cv\ (k)\}.$$

Note that $(u,k) = 1$ automatically. Write $\Lambda = \log(z/H)$.

Lemma 3.1. *There is an absolute constant K_0 such that the following holds. Let $K \ge K_0$, $k \le K$, $(c,k) = 1$, $H = K^{10}$ and $z \ge H^2$. Then*

$$\sum_{(u,v) \in \mathcal{S}(k,c;H,z)} \frac{1}{uv} \ \ge\ \frac{1}{4}\,\frac{\varphi(k)}{k^2}\,\Lambda^2, \tag{3}$$

$$\sum_{(u,v) \in \mathcal{S}(k,c;H,z)} \frac{1}{\varphi(u)\varphi(v)} \ \le\ C_1\,\frac{\varphi(k)}{k^2}\,\Lambda^2 \tag{4}$$

with an absolute constant C_1.

Proof. We use two elementary estimates, valid for $1 \le H' < z'$, $\Lambda' = \log(z'/H')$, $(a,k) = 1$:

$$\sum_{\substack{H' < u \le z' \\ u \equiv a\ (k)}} \frac{1}{u} = \frac{\Lambda'}{k} + O\Big(\frac{1}{H'}\Big), \tag{5}$$

$$\sum_{\substack{H' < v \le z' \\ (v,k) = 1}} \frac{1}{v} = \frac{\varphi(k)}{k}\Lambda' + O\Big(\frac{\tau(k)}{H'}\Big). \tag{6}$$

For (5), the terms are $u_0 + jk$ with $H' < u_0 \le H' + k$, and comparison with $\frac{1}{k}\int dt/t$ gives the sum within $1/H'$ of Λ'/k (for a nonempty sum the upper bound is $1/u_0 + \frac{1}{k}\log(z'/u_0)$; the lower bound is $\frac{1}{k}\log(z'/(H'+k)) \ge \Lambda'/k - 1/H'$). For an empty sum, $z' < u_0 \le H' + k$ gives $\Lambda'/k \le 1/H'$ directly. For (6), Möbius inversion over $e \mid k$ and $\sum_{A < m \le B} 1/m = \log(B/A) + O(1/A)$ give $\sum_{e \mid k} \frac{\mu(e)}{e}\big(\Lambda' + O(e/H')\big)$.

Lower bound. Let $T(H', z')$ denote the sum of $1/(uv)$ over $H' < u, v \le z'$ with $(v, k) = 1$ and $u \equiv -cv \pmod k$ (no coprimality of u, v). Summing (5) over v and then using (6),

$$T(H', z') = \frac{\varphi(k)}{k^2}\Lambda'^2 + O\Big(\frac{\Lambda'\tau(k)}{H'}\Big).$$

valid for $H' \ge 1$ and $\Lambda' \ge 1$: the additional product error $O(\tau(k)/H'^2)$ is then absorbed. Möbius inversion over $d = (u, v)$ (necessarily $(d, k) = 1$, and d is invertible modulo k, so the congruence descends to $u' \equiv -cv'$) gives

$$\sum_{\mathcal{S}} \frac{1}{uv} = \sum_{\substack{d \le z \\ (d,k)=1}} \frac{\mu(d)}{d^2}\, T(H/d, z/d),$$

and $\log\big((z/d)/(H/d)\big) = \Lambda$ for every d. For $d > H$ we use only the trivial bound $T(H/d, z/d) \le (1+\log z)^2$ (each one-variable sum is at most $1+\log z$), which contributes at most $(1+\log z)^2 \sum_{d>H} d^{-2} \ll \Lambda^2/H$ in total, using $\log z \le 2\Lambda$ (from $z \ge H^2$). Hence

$$\sum_{\mathcal{S}} \frac{1}{uv} = \frac{\varphi(k)}{k^2}\Lambda^2 \sum_{\substack{d \le H \\ (d,k)=1}} \frac{\mu(d)}{d^2} + O\Big(\frac{\Lambda\tau(k)\log z}{H} + \frac{\Lambda^2}{H}\Big).$$

The sum over d is $\prod_{p\nmid k}(1-p^{-2}) + O(1/H) \ge 6/\pi^2 - O(1/H)$. Since $\tau(k)k^2/\varphi(k) \le k\tau(k)^2 \le 4K^2$ (using $k/\varphi(k) \le 2^{\omega(k)} \le \tau(k) \le 2\sqrt{k}$), the whole error is at most $CK^2H^{-1} \cdot \frac{\varphi(k)}{k^2}\Lambda^2 = CK^{-8}\,\frac{\varphi(k)}{k^2}\Lambda^2$ for an absolute C. This gives (3) for $K \ge K_0$, with an absolute K_0; below we only use $K \to \infty$, so we assume $K \ge K_0$ from now on.

Upper bound. Drop $(u, v) = 1$ and use $1/\varphi(m) = \frac{1}{m}\sum_{d\mid m} \mu^2(d)/\varphi(d)$. For $(a, k) = 1$, only d with $(d, k) = 1$ contribute. By (5) for $d \le H$, and by the trivial bound $1 + \log z \le 1 + 2\Lambda$ for $d > H$,

$$\begin{aligned}
\sum_{\substack{H < u \le z \\ u \equiv a\,(k)}} \frac{1}{\varphi(u)} &= \sum_{\substack{d \le z \\ (d,k)=1}} \frac{\mu^2(d)}{d\varphi(d)} \sum_{\substack{H/d < u' \le z/d \\ u' \equiv ad^{-1}\,(k)}} \frac{1}{u'} \\
&\le \frac{\Lambda}{k}\sum_{d} \frac{\mu^2(d)}{d\varphi(d)} + \frac{1}{H}\sum_{d \le H} \frac{\mu^2(d)}{\varphi(d)} + (1+2\Lambda)\sum_{d>H} \frac{1}{d\varphi(d)} \\
&\le C\Big(\frac{\Lambda}{k} + \frac{\log H}{H} + \frac{\Lambda}{\sqrt{H}}\Big),
\end{aligned}$$

because

$$\sum_{d} \frac{\mu^2(d)}{d\varphi(d)} < \infty, \qquad \sum_{d \le H} \frac{\mu^2(d)}{\varphi(d)} \ll \log H, \qquad \sum_{d>H} \frac{1}{d\varphi(d)} \ll H^{-1/2}.$$

Likewise, by (6),

$$\sum_{\substack{H<v\le z\\(v,k)=1}}\frac{1}{\varphi(v)}\le C\Big(\frac{\varphi(k)}{k}\Lambda+\frac{\tau(k)\log H}{H}+\frac{\Lambda}{\sqrt{H}}\Big).$$

Since $k\le K$, $\tau(k)\le 2\sqrt{K}$, $\varphi(k)/k\ge 1/\tau(k)\ge 1/(2\sqrt{K})$, $H=K^{10}$ and $\Lambda\ge\frac12\log z\ge 10\log K$, the secondary terms in both brackets are $O(K^{-4})$ times the main terms. Multiplying proves (4). □

Lemma 3.2. *$h(K)\asymp\log K$ for $K\ge 5$.*

Proof. Writing $\varphi(k)/k=\sum_{d\mid k}\mu(d)/d$,

$$h(K)=\sum_{\substack{d\le K\\ d\text{ odd}}}\frac{\mu(d)}{d^2}\sum_{\substack{m\le K/d\\ m\equiv d^{-1}\,(4)}}\frac{1}{m}=\sum_{\substack{d\le K\\ d\text{ odd}}}\frac{\mu(d)}{d^2}\Big(\frac14\log\frac{K}{d}+O(1)\Big)$$

and the last expression equals $\frac{2}{\pi^2}\log K+O(1)$, because $\sum_{d\,\mathrm{odd}}\mu(d)d^{-2}=\prod_{p>2}(1-p^{-2})=8/\pi^2$ and $\sum_d\mu^2(d)d^{-2}(1+\log d)<\infty$. □

4. The class-mass lemma

Given X, fix the blocks $(x_j,2x_j]$ with $x_j=2^jX^{1/2}$ and $0\le j<J=\lceil\log X/(2\log 2)\rceil$; the last may extend beyond X. Fix $K\ge K_0$, put $H=K^{10}$, and fix c with $(c,L_K)=1$. For a prime $\ell\equiv 3\pmod 4$ lying in one of these blocks $(x,2x]$, put $z=x^{1/6}$ and let $\Omega_c(\ell)$ be the set of residues $-uv^{-1}\pmod\ell$ arising from triples

$$k\in\mathcal{K}(K),\qquad (u,v)\in\mathcal{S}(k,c;H,z),\qquad 4uv\mid k\ell+1. \tag{7}$$

By Lemma 2.1 and (2) (with $w=(k\ell+1)/(4uv)$), every $n\equiv c\pmod{L_K}$ with $n\bmod\ell\in\Omega_c(\ell)$ satisfies (1). Let $\omega_c(\ell)=|\Omega_c(\ell)|$, and set $\Omega_c(\ell)=\varnothing$, $\omega_c(\ell)=0$ for primes $\ell\not\equiv 3\pmod 4$.

Lemma 4.1. *There are absolute constants $a,A>0$ and X_0 such that the following holds. Let $X\ge X_0$, $K_0\le K\le(\log X)^4$ (with K_0 from Lemma 3.1), $H=K^{10}$, $(c,L_K)=1$. Then, for every prime $\ell\in(X^{1/2},X]$, $\omega_c(\ell)\le\ell^{1/3}$, and*

$$a(\log X)^2h(K)\ \le\sum_{\substack{X^{1/2}<\ell\le X\\ \ell\equiv 3\,(4)}}\frac{\omega_c(\ell)}{\ell}\ \le\ A(\log X)^2h(K). \tag{8}$$

Proof. Distinctness. Fix $\ell\in(x,2x]$. Two triples (k,u,v), (k',u',v') in (7) with the same residue satisfy $\ell\mid uv'-u'v$ and $|uv'-u'v|<z^2=x^{1/3}<\ell$, so $uv'=u'v$ and, both pairs being coprime, $(u,v)=(u',v')$. Then $k\ell\equiv k'\ell\equiv-1\pmod{4uv}$ forces $k\equiv k'\pmod{4uv}$, and $4uv>4H^2>K$ gives $k=k'$. Hence $\omega_c(\ell)$ equals the number of triples (7), and $\omega_c(\ell)\le z^2\le\ell^{1/3}$.

Dyadic blocks. Use the fixed blocks above, taking the last block in full for the upper bound and discarding it for the lower bound if it extends beyond X. There are $\asymp\log X$ blocks in either case, and $\log x\asymp\log X$ in each.

Since $K \le (\log X)^4$ and X is large, $z = x^{1/6} \ge H^2 = K^{20}$, so Lemma 3.1 applies with $\Lambda = \log(z/H) \ge \frac{1}{2}\log z = \frac{1}{12}\log x$. For a triple, the condition $4uv \mid k\ell + 1$ is the reduced progression

$$\ell \equiv -k^{-1} \pmod{4uv}, \qquad 4uv \le 4x^{1/3}, \tag{9}$$

which already implies $\ell \equiv 3 \pmod 4$. Thus

$$\sum_{x<\ell\le 2x} \omega_c(\ell) = \sum_{k\in\mathcal{K}(K)} \sum_{(u,v)\in\mathcal{S}(k,c;H,z)} \big(\pi(2x; 4uv, -k^{-1}) - \pi(x; 4uv, -k^{-1})\big). \tag{10}$$

Upper bound. By Brun–Titchmarsh, $\pi(2x; q, a) \ll x/(\varphi(q)\log x)$ for $q \le x^{1/2}$, and $\varphi(4uv) \ge 2\varphi(u)\varphi(v)$; so by (4)

$$\sum_{x<\ell\le 2x} \omega_c(\ell) \ll \frac{x}{\log x} \sum_k \sum_{\mathcal{S}} \frac{1}{\varphi(u)\varphi(v)} \ll \frac{x}{\log x}\,(\log x)^2\, h(K) = x\log x\, h(K).$$

Dividing by x and summing over the blocks gives the upper half of (8).

Lower bound. Write $E(x; q, a) = \pi(2x; q, a) - \pi(x; q, a) - (\mathrm{li}(2x) - \mathrm{li}(x))/\varphi(q)$ and $E^*(q) = \max_{(a,q)=1} |E(x; q, a)|$. The main term of (10) is, by $\mathrm{li}(2x) - \mathrm{li}(x) \ge x/\log(2x)$, $1/\varphi(4uv) \ge 1/(4uv)$ and (3),

$$\ge \frac{x}{4\log 2x} \sum_{k\in\mathcal{K}(K)} \frac{1}{4}\frac{\varphi(k)}{k^2}\Lambda^2 \;\ge\; c_0\, x\log x\, h(K)$$

for an absolute $c_0 > 0$. For the error, group the triples by $q = 4uv$: a given q arises from at most $K\tau(q)$ triples (at most K values of k and $\tau(q/4)$ ordered factorizations $uv = q/4$). By Cauchy–Schwarz, the trivial bound $E^*(q) \ll x/(\varphi(q)\log x)$ (Brun–Titchmarsh), the estimate $\sum_{q\le Q} \tau(q)^2/\varphi(q) \ll (\log Q)^4$, and the Bombieri–Vinogradov theorem in the form $\sum_{q\le x^{1/2}(\log x)^{-B}} E^*(q) \ll_R x(\log x)^{-R}$, where $B = B(R)$ (note $4x^{1/3}$ lies below this level for large x),

$$\begin{aligned}\sum_{q\le 4x^{1/3}} K\tau(q)E^*(q) &\le K\Big(\sum_q \tau(q)^2 E^*(q)\Big)^{1/2}\Big(\sum_q E^*(q)\Big)^{1/2} \\ &\ll K\big(x(\log x)^3 \cdot x(\log x)^{-R}\big)^{1/2} \le \frac{x}{(\log x)^3}\end{aligned}$$

on taking $R = 19$ and using $K \le (\log X)^4 \le 32(\log x)^4$. Since $h(K) \ge 1$ (the term $k = 1$), the error is $o(x\log x\, h(K))$ and

$$\sum_{x<\ell\le 2x} \frac{\omega_c(\ell)}{\ell} \;\ge\; \frac{1}{2x} \sum_{x<\ell\le 2x} \omega_c(\ell) \;\ge\; \frac{c_0}{3}\log x\, h(K).$$

Summing over the $\asymp \log X$ full blocks contained in $(X^{1/2}, X]$ gives the lower half of (8). For completeness, the divisor estimate used above follows by bounding its sum by the Euler product with local factors $1 + \sum_{j\ge 1}(j+1)^2/\varphi(p^j) = 1 + 4/p + O(p^{-2})$ and applying Mertens' bound. The quoted Bombieri–Vinogradov form follows from the standard version with $\max_{(a,q)=1} \max_{2\le y\le 2x} |\pi(y; q, a) - \mathrm{li}(y)/\varphi(q)|$; the two endpoints cost

only a factor 2. The bound for $E^*(q)$ includes the main term, since $\mathrm{li}(2x) - \mathrm{li}(x) \ll x/\log x$. $\square$

The point of Lemma 4.1 is its uniformity in c: the only place c entered was Lemma 3.1, whose bounds hold for every reduced residue. Without multipliers (only $k = 1$) the argument is Lemma 4.1 of [5].

5. The large sieve; proof of Theorem 1.1

We use Montgomery's large sieve [3, §8, p. 561] in the following form. Let $\mathcal{N}$ be a set of integers in an interval of length Y, and let $\mathcal{P}$ be a finite set of primes such that, for each $\ell \in \mathcal{P}$, the elements of $\mathcal{N}$ avoid $0 \le \omega(\ell) < \ell$ residue classes modulo ℓ. Then for every $Q \ge 1$

$$|\mathcal{N}| \le \frac{Y + Q^2}{S(Q)}, \qquad S(Q) = \sum_{\substack{s \le Q \\ s \mid \prod \mathcal{P}}} \mu^2(s) \prod_{\ell \mid s} g(\ell), \qquad g(\ell) = \frac{\omega(\ell)}{\ell - \omega(\ell)}. \tag{11}$$

We also use the standard lower bound for $S(Q)$ (as in [5, §4]). Put $G = \prod_{\ell \in \mathcal{P}} (1 + g(\ell)) = \sum_{s \mid \prod \mathcal{P}} \mu^2(s) \prod_{\ell \mid s} g(\ell)$. For any $v > 0$, Rankin's trick gives

$$\frac{G - S(Q)}{G} \le Q^{-v} \prod_{\ell \in \mathcal{P}} \frac{1 + g(\ell)\ell^v}{1 + g(\ell)} \le Q^{-v} \exp\Big\{ \sum_{\ell \in \mathcal{P}} g(\ell)(\ell^v - 1) \Big\}.$$

If $\mathcal{P} \subseteq (1, X]$ and $v = 1/\log X$, then $\ell^v - 1 \le \mathrm{e} - 1$, and if moreover $\omega(\ell) \le \ell/2$ then $g(\ell) \le 2\omega(\ell)/\ell$; hence

$$\frac{G - S(Q)}{G} \le \exp\Big\{ -\frac{\log Q}{\log X} + 2(\mathrm{e} - 1) \sum_{\ell \in \mathcal{P}} \frac{\omega(\ell)}{\ell} \Big\}. \tag{12}$$

Proof of Theorem 1.1. Let N be large, fix an absolute $0 < \delta < 1/4$, and put $K = \lfloor \delta \log N \rfloor$, $M = L_K$. Since $L_K \mid \mathrm{lcm}(1, \dots, K) = \mathrm{e}^{\psi(K)}$ and $\psi(K) \le 2K$ (Chebyshev), $M \le N^{2\delta}$, so $Y := N/M \ge N^{1-2\delta}$. Put

$$X = \exp\Big\{ \alpha \Big(\frac{\log N}{\log\log N} \Big)^{1/3} \Big\} \tag{13}$$

with a small absolute $\alpha > 0$ to be chosen; then $K \le (\log X)^4$ for large N, so Lemma 4.1 applies.

Every prime $p > K$ is coprime to M. Fix a reduced representative $1 \le c \le M$ and let $\mathcal{N}_c$ be the set of integers $m \in [0, Y]$ for which $c + Mm$ is an exceptional prime not exceeding N. For each prime $\ell \in (X^{1/2}, X]$, $\ell \equiv 3 \pmod 4$, we have $\ell \nmid M$ (as $\ell > K$), and by Lemma 2.1 the number $n = c + Mm$ is solvable whenever $n \bmod \ell \in \Omega_c(\ell)$; so m avoids the $\omega_c(\ell)$ classes $(\rho - c)M^{-1} \pmod \ell$, $\rho \in \Omega_c(\ell)$. Apply (11) with $Q = Y^{1/2}$: $|\mathcal{N}_c| \le 2Y/S(Q)$. By Lemma 4.1, $\omega_c(\ell) \le \ell^{1/3} \le \ell/2$ and $\sum_\ell \omega_c(\ell)/\ell \le A(\log X)^2 h(K) \le A'(\log X)^2 \log K$ (Lemma 3.2); so (12) gives

$$\frac{G - S(Q)}{G} \le \exp\Big\{ -\frac{(1 - 2\delta)\log N}{2\log X} + 2(\mathrm{e} - 1)A'(\log X)^2 \log K \Big\} \le \frac{1}{2}$$

provided α is so small that $(\log X)^3 \log K \le (\log N)/(16(\mathrm{e}-1)A')$; by (13) and $\log K \le \log\log N$ this holds for all large N. Therefore, using $\log(1+g) \ge g/2 \ge \omega/(2\ell)$ and the lower half of (8),

$$S(Q) \ge \frac{G}{2} \ge \frac{1}{2}\exp\Big\{\frac{1}{2}\sum_{\ell}\frac{\omega_c(\ell)}{\ell}\Big\} \ge \frac{1}{2}\mathrm{e}^{a(\log X)^2 h(K)/2} \ge \frac{1}{2}\mathrm{e}^{a'(\log X)^2\log K},$$

uniformly in c. Summing over the $\varphi(M) \le M$ reduced classes and adding the primes $p \le K$,

$$E_{\mathrm{pr}}(N) \le K + M\cdot\frac{4Y}{\mathrm{e}^{a'(\log X)^2\log K}} \ll N\exp\Big\{-a'\alpha^2\,\frac{(\log N)^{2/3}\log(\delta\log N)}{(\log\log N)^{2/3}}\Big\},$$

which is the theorem. □

6. All denominators; proof of Theorem 1.2

If $p \mid n$ and $4/p = 1/x_1 + 1/x_2 + 1/x_3$ then $4/n = \sum_i 1/((n/p)x_i)$. Hence every exceptional $n > 1$ has all its prime factors exceptional, i.e. lies in the multiplicative semigroup $\mathcal{S}$ generated by the exceptional primes, including the empty product 1. Put $g(u) = u^{2/3}(\log u)^{1/3}$ and choose $c_0 > 0$ small enough that Theorem 1.1 gives $E_{\mathrm{pr}}(t) \le t\exp\{-c_0 g(\log t)\}$ for $t \ge t_0 \ge \exp(\mathrm{e})$. Fix $0 < \eta < c_0$, let x be large, $L = \log x$, and $\delta = \eta g(L)/L \in (0, 1/4)$. By Rankin's trick,

$$E(x) \le \sum_{\substack{n\in\mathcal{S}\\ n\le x}}\Big(\frac{x}{n}\Big)^{1-\delta} \le x^{1-\delta}\prod_{\substack{p\le x\\ p\ \text{exceptional}}}(1-p^{-1+\delta})^{-1}.$$

The product is $\ll \exp\{\sum_{p\le x\ \text{exc.}} p^{-1+\delta}\}$, since higher prime powers contribute $O(\sum_p p^{-3/2}) = O(1)$ uniformly for $\delta < 1/4$. The endpoint in partial summation is $E_{\mathrm{pr}}(x)x^{-1+\delta} \le \exp\{-(c_0-\eta)g(L)\} \le 1$; hence

$$\sum_{\substack{t_0<p\le x\\ p\ \text{exc.}}} p^{-1+\delta} \ll 1 + \int_{t_0}^{x} E_{\mathrm{pr}}(t)\,t^{-2+\delta}\,dt \le 1 + \int_{\log t_0}^{L}\exp\{\delta u - c_0 g(u)\}\,du.$$

Since $g(u)/u$ is decreasing for $u \ge \mathrm{e}$, we have $\delta u \le \eta g(u)$ for $\log t_0 \le u \le L$, so the integral is at most $\int_{\mathrm{e}}^{\infty}\exp\{-(c_0-\eta)g(u)\}du < \infty$. Thus the Euler product is bounded uniformly in x, and $E(x) \ll x^{1-\delta} = x\exp\{-\eta g(\log x)\}$. □

7. Remarks

Remark 7.1 (general numerators)**.** *Replacing* 4 *by a fixed* $m \ge 4$ *(the Erdős–Straus–Schinzel problem* $m/n = 1/x + 1/y + 1/z$*), the identity of Lemma 2.1 becomes: if* $k\ell \equiv -1 \pmod m$*,* $A = (k\ell+1)/m = uvw$ *and* $nv \equiv -u \pmod{k\ell}$*, then* $m/n = 1/(suw) + 1/(nsvw) + 1/(nuvw)$*, where* $s = (nv+u)/(k\ell)$*. Take* $k \equiv 1 \pmod m$*,* $\ell \equiv -1 \pmod m$*, and replace* $4uv$ *by* muv *in the class count. The lattice lemma is unchanged, while* $\sum_{k\le K,\ k\equiv 1\,(m)}\varphi(k)/k^2 \asymp_m \log K$ *by the proof of Lemma 3.2; all remaining*

constants may depend on m. The argument above then gives $E_m(N) \ll_m N \exp\{-c_m(\log N)^{2/3}(\log\log N)^{1/3}\}$; we have not tracked the dependence on m, which [5] *make explicit for Vaughan's bound.*

Remark 7.2 (on the method)**.** *Vaughan's argument is a large sieve with prime moduli ℓ and, on average, $\asymp (\log \ell)^2$ excluded classes per modulus. Lemma 4.1 raises the class mass from $(\log X)^2$ to $(\log X)^2 h(K)$ in its stated range. Extrapolating formally to $\log K \asymp \log X$ suggests a cubic class mass, but this is not established here. Nor do the prime-average solution counts of* [1] *establish a typical order $\asymp (\log n)^3$. The present note uses a single progression modulo $L_K \le N^{1/2}$ to absorb the k-part, which produces the $\log\log$ factor. A cubic class mass with comparable sieve and tail control would give an exponent $3/4$; obtaining such control is not addressed here. The bound $E(N) \ll N \exp\{-c(\log N)^{3/4}\}$, which supersedes the theorem of this note, was obtained by Kandukoori, Ramesh and Kumar* [2] *(unrefereed preprint dated 29 July 2026, with a Lean formalisation) and, internally reviewed only, in the companion note* `paper/es-threequarter-note.tex`*.*

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